REWARD ERRORS CHANGE MEMORY BEFORE THEY CHANGE POLICY

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ABSTRACT

Self-evolving agents often turn terminal self-evaluation into persistent memory, so a wrong label can become a state update rather than merely a scoring error. We study where such a label-only intervention survives along the write→retrieval→policy-uptake→outcome path in released ReasoningBank/WebArena. Holding each source experience byte-identical, success/failure reflection produces different durable memories on 20/20 complete Shopping pairs; reward-mode divergence also exceeds a stronger same-mode wording control (0.700 vs. 0.595; $p = 0.0078$). Forced memory injection shows latent leverage (terminal $|\Delta| = 0.15625$, $p = 0.00074$), but native Shopping transport is weak: 36 retrieval-hit tasks yield terminal $|\Delta| = 0.02083$ ($p = 0.4289$), first-action TV 0.06944 ($p = 0.5801$), and 0/36 modal branch differences. An outcome-blind structured-memory control also misses the 0.15 terminal floor (0.04514, $p = 0.0048$). A separately qualified Reddit replication changes all four required source-memory pairs, yet native terminal $|\Delta| = 0.125$ ($p = 0.2253$) still misses the same floor, with 6/8 zero tasks. Thus reward labels robustly change stored state, while exposure, branch-specific uptake, and realized magnitude are distinct and domain-dependent gates.

1 INTRODUCTION

Agents with external memory do more than score completed episodes: they can turn the score into state for future decisions. In ReasoningBank-style systems, a trajectory is judged, distilled through a success- or failure-conditioned reflection objective, stored, retrieved for a later task, and injected into the policy context. If the terminal label is wrong, the same experience can therefore be committed through a different update rule without any parameter change. This makes evaluator reliability a persistent-state reliability problem—but only if the resulting state difference survives retrieval and policy use.

Prior work already establishes the surrounding pieces. JudgeBench and RewardBench study evaluator reliability (Tan et al., 2025; Lambert et al., 2024); ReasoningBank and Reflexion convert feedback into reusable textual memory (Ouyang et al., 2026; Shinn et al., 2023); Mem-α and AttriMem learn memory construction from outcome or process reward (Wang et al., 2025; Li et al., 2026a); and Memory Reward Inflation, RoMeRL, and recent fragility work analyze reward-driven reuse or system-level self-improvement instability (Asadolahi et al., 2026; Yang et al., 2026; Ye et al., 2026). We therefore do not claim that judges fail, that feedback shapes memory, or that memory affects later behavior. The missing object is stage-resolved: *for a byte-identical experience, where does a terminal-label intervention survive or attenuate along the path from writing durable state to native exposure, branch-specific policy uptake, and eventual outcome?*

This distinction matters because two common evaluations answer different questions. A forced memory swap asks whether two persistent states *can* change behavior when directly injected. A native test asks whether the released system actually retrieves one of those states for a future task and whether the policy uses the success/failure distinction strongly enough to change decisions. Conflating the two can turn a valid intervention result into an unsupported deployment claim.

We first identify the write treatment. Task and trajectory bytes are fixed while only the released success- versus failure-reflection mode changes. On the original complete pairs and a preregistered 16-source breadth replication, all 20 complete pairs change both memory content and title sets (pooled token-set Jaccard distance 0.673). Because the two reflection branches necessarily use different prompts, we challenge the effect with a stronger same-mode semantic-preserving wording perturbation on eight disjoint trajectories. Reward-mode divergence is 0.700 versus 0.595 for wording alone, a paired excess of 0.105 ($p = 0.0078$). Thus the write-stage result is not reduced to generic prompt sensitivity.

We then trace the intervention through the remaining gates. A fully crossed forced 4×4 terminal swap passes the frozen practical and statistical criteria (mean $|\Delta| = 0.15625$, $p = 0.00074$), establishing latent conditional leverage. Exact replay of the released top-1/0.3 retriever, however, shows that none of those four futures retrieves an original-bank memory. Expanding the bank to 20 sources raises held-out Shopping exposure to 125/172 tasks; before terminal outcomes we freeze all 36 retrieval hits with reconstructable released evidence and deterministic evaluation. Native success/failure terminal separation then falls to 0.02083 ($p = 0.4289$), with 34/36 tasks showing no branch contrast.

The attenuation is already visible before terminal scoring. On the same 36 native-hit states, a preregistered first-action probe gives mean success/failure action-distribution TV 0.06944 ($p = 0.5801$) against a 0.20 process floor, and the modal action differs in 0/36 states. We also attack simpler representational explanations. Literal omission and the raw writer-input trajectory each produce only 0.04514 mean absolute terminal separation from the reward-conditioned memories. More strongly, we generate an outcome-blind procedural memory from the *same task and action-summary bytes*, with the same writer family and Title/Description/Content schema but no success/failure, reward, score, or outcome semantics. All 20 neutral memories complete and form distinct textual states; on the same 36 native tasks, reward-conditioned memories differ from this structured control by only 0.04514 on average ($p = 0.0048$), below the unchanged 0.15 practical floor. Thirty-two tasks have zero reward-conditioned-versus-neutral effect and 30/36 have identical success, failure, neutral, raw, and no-memory point estimates.

A second-domain replication tests whether the Shopping magnitude is itself the story. Before any new writer call, a zero-call Reddit qualification freezes 20 outcome-balanced source descriptions and finds eight deterministic retrieval-hit futures across two intent templates and four selected source identities. All four required success/failure memory pairs diverge (mean token Jaccard 0.652). On the frozen eight-task native support, mean terminal $|\Delta| = 0.125$ with $p = 0.2253$, again below the unchanged 0.15 floor; six tasks are zero and the two nonzero cells have opposite signs. Thus write divergence transfers more cleanly than native magnitude, while the amount of attenuation is domain/task dependent.

Contribution and boundary. The contribution is a stage-resolved causal decomposition of reward-conditioned agent memory. First, a label-only write intervention plus a stronger wording reduction identifies robust durable-state change. Second, forced injection establishes latent leverage without being mislabeled as native transport. Third, exact retrieval, first-action uptake, raw-experience, outcome-blind structured-memory, endpoint-resolution, and second-domain controls show that broad native attenuation is real but its magnitude is domain/task dependent. We make no claim of a new memory architecture, a new reward-learning method, universally harmful failure memory, a confirmatory generic memory-presence effect, writer- or policy-invariant downstream effects, live-browser transport, or a population law. The supported lesson is narrower: memory-text divergence is evidence of state corruption, not by itself evidence of realized behavioral transport.

2 Related Work

Evaluator reliability before adaptation. JudgeBench, RewardBench, and Plan-RewardBench expose failures of LLM judges and reward models on objectively or procedurally checkable inputs (Tan et al., 2025; Lambert et al., 2024; Wang et al., 2026a). They establish that a terminal signal can be wrong. Our question begins one step later: what happens when that signal chooses a persistent-state update and the resulting state is reused?

Feedback-conditioned memory construction. ReasoningBank distills structured reasoning memories from self-judged successful and failed experiences, while Reflexion stores evaluative verbal feedback across trials (Ouyang et al., 2026; Shinn et al., 2023). Mem-α learns what to store from downstream reward, and AttriMem adds attribution-derived process feedback to address coarse outcome-level credit assignment (Wang et al., 2025; Li et al., 2026a). These works make feedback-

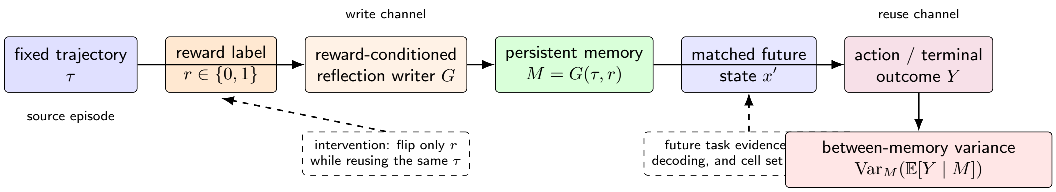

Figure 1: Stage-resolved persistent-state channel. A terminal label selects the write objective for a fixed source trajectory. Forced injection directly tests latent leverage of the resulting state; native transport additionally requires retrieval exposure and branch-specific policy uptake before a terminal effect can occur.

conditioned construction established prior art. We instead freeze the construction substrate and source experience, use the terminal label as the sole treatment, and ask where that intervention remains distinguishable after storage.

Memory representation, retrieval, and reuse. Raw trajectories and long-context baselines remain important competitors for agent memory: ReasoningBank explicitly compares against raw experience, while ALMA and EvoMemBench evaluate learned or designed memories against trajectory-retrieval and strong long-context controls (Ouyang et al., 2026; Xiong et al., 2026; Wang et al., 2026b). Very recent work further separates retrieval from post-retrieval reuse by holding the retrieved candidate and target state fixed while changing the support delivered to the policy (Li et al., 2026b). We therefore do not compare reward-conditioned memory only with omission, nor treat retrieval success as realized use. Our controls include the exact compact writer-input trajectory and an outcome-blind structured procedural rewrite using the same information and memory schema.

Reward-driven memory reuse and self-improvement variability. Memory Reward Inflation studies how proxy reward can overvalue stored memories and amplify reuse; RoMeRL targets misleading utility updates and the memory-reward trap; Ye et al. quantify variance, task-order sensitivity, and underspecification in self-improving agents (Asadolahi et al., 2026; Yang et al., 2026; Ye et al., 2026). Their interventions operate at valuation, utility update, retrieval/reuse, or whole-system variation. Our treatment is earlier and more local: a controlled write-time label switch followed by a stage-resolved audit of exposure, policy uptake, and outcome.

Defended residual. The residual claim is therefore neither “reward can be wrong” nor “feedback changes memory.” It is that a byte-identical label intervention can be identified at the durable-state boundary, while downstream evidence must separately establish whether that difference is exposed and converted into branch-specific policy behavior. This separation lets positive forced-intervention evidence coexist with a negative native-transport boundary without contradiction.

3 Causal Object: Where Does a Label Intervention Survive?

Let source task x produce trajectory τ, and let a terminal evaluator emit $r \in \{0, 1\}$. A fixed reward-conditioned writer constructs persistent memory

$$M_r = G(\tau, r). \tag{1}$$

In released ReasoningBank, $r = 1$ and $r = 0$ select success- and failure-reflection objectives. Our write-stage intervention holds x and τ fixed and changes only this branch. The first gate is therefore *state construction*: does $M_1 \neq M_0$ under a label-only treatment, beyond ordinary prompt sensitivity?

For a future query x', the released retriever chooses a source identity from task-description embeddings. Let $E(x') \in \{0, 1\}$ indicate that a source memory is exposed to the policy. Because reward-conditioned memory content is not used in this retrieval embedding, exposure is a separate gate from the success/failure branch itself. Conditional on exposure, the downstream policy produces an action distribution

$$A \sim \pi(\cdot \mid o, M_r), \tag{2}$$

where o is the held-fixed future evidence. We call the S/F difference in this distribution *branch-specific uptake*. A terminal outcome Y is then a later consequence of the resulting interaction. The native chain is thus

$$\text{label } r \to M_r \to E(x') \to A \to Y. \tag{3}$$

A forced memory swap uses $do(M = M_r)$ and therefore bypasses the exposure gate. It answers whether a state difference can have leverage under direct injection, not whether the released system will realize that leverage broadly.

To distinguish branch-specific semantics from generic rewriting, we also construct an outcome-blind structured control

$$M_N = G_N(\tau), \tag{4}$$

using the same task and action-summary information, writer family, and memory schema but no success/failure, reward, score, or outcome semantics. On the frozen native support, the preregistered terminal statistic is

$$D_N = \frac{1}{|\mathcal{T}|} \sum_{x' \in \mathcal{T}} \frac{|p_1(x') - p_N(x')| + |p_0(x') - p_N(x')|}{2}, \tag{5}$$

where p_r and p_N are conditional success rates. A practically large reward-conditioned effect beyond generic structured rewriting requires both $D_N \geq 0.15$ and the frozen permutation criterion $p < 0.05$.

3.1 Bounded variance consequence

If label corruption stochastically selects between two measured persistent states, the law of total variance gives

$$\mathrm{Var}(Y) = \mathbb{E}_M[\mathrm{Var}(Y \mid M)] + \mathrm{Var}_M(\mathbb{E}[Y \mid M]). \tag{6}$$

The second term is nonzero when the selected memory changes conditional outcomes. We use this only as a consequence analysis of the forced intervention. It is not evidence that native retrieval, uptake, or population risk is large.

4 Experimental Setup

Write-stage identification. We use released WebArena Shopping trajectories from the ReasoningBank/fragility substrate. The original intervention requests six outcome-balanced source trajectories; four return complete memory outputs under both reward branches and two failure-branch calls terminate before assistant text. We do not impute or re-POST these missing arms. A disjoint eight-trajectory prompt control compares released reward-mode divergence with a larger semantic-preserving rewording within each mode. For breadth, 16 additional sources are deterministically frozen before new writer outputs, balanced 8/8 by original outcome and spanning 11 intent-template families. A uniform 4,096-token execution repair regenerates all 32 breadth writer units fresh after the lower-cap parent reveals length censoring. Combined with the original complete sample, this yields 20 paired sources.

Four transport gates. We evaluate state construction, exposure, branch-specific uptake, and terminal outcome separately. Forced injection fully crosses the four original complete memory pairs with four frozen future evidence packets; this bypasses native retrieval and tests latent leverage. Exposure is measured by exact replay of the released all-MiniLM-L6-v2 task-description retriever with top-1 and threshold 0.3. After source expansion, all held-out Shopping retrieval hits are screened only for released evidence and deterministic evaluation, producing 36 native tasks before terminal outcomes. Each is bound to the source selected by the retriever itself.

Native outcome and representation controls. On the 36-task support, success- and failure-conditioned memories use exact stored bytes and the released ReasoningBank human-memory wrapper; future evidence, Doubao Seed 2.0 Mini policy, temperature, evaluator, and four-rollout depth are matched. A no-memory arm uses the same support. A raw writer-input arm substitutes the deterministic compact action/result trace supplied to the memory writer before reward conditioning. Terminal practical-effect tests retain the frozen 0.15 floor together with permutation $p < 0.05$.

The strongest same-information representation control is an outcome-blind structured writer. For all 20 sources, DeepSeek-V4-Flash receives the same task prompt and exact action-summary bytes used by the reward-conditioned evidence, at temperature zero and a 4,096-token cap. The control prompt uses the same at-most-three Title/Description/Content schema but contains no success/failure, reward, score, or outcome semantics. All 20 writer calls are frozen before execution; no failed unit may be retried, replaced, or imputed. The 11 neutral memories needed by native retrieval are then serialized with the same wrapper. On the same 36 tasks, 144 fresh terminal calls estimate Eq. 5; both the 0.15 magnitude floor and $p < 0.05$ are required.

Pre-terminal uptake. To localize attenuation before terminal scoring, the first-action probe freezes released trajectory step 1 for all 36 native tasks, removes any pre-existing task-history memory from the state message, and compares success-memory, failure-memory, and no-memory first structured actions with four rollouts each. The preregistered statistic is mean per-state empirical S/F total-variation distance, using the inherited 0.20 process floor and a 100,000-draw within-state permutation test. No-memory geometry and later next-goal text are descriptive only.

Reliability and scope. All scientific runs are inference-only: no training and no local GPU fine-tuning. Provider responses are archived before analysis where supported; each outcome-blind writer/terminal provider POST additionally writes a resumable stage JSON and refreshes the aggregate result. Missing scientific units are never replaced by outcome-favorable tasks. A DeepSeek-V4-Flash downstream-policy transfer stops after no-text length censoring on the first frozen unit at both allowed output caps and therefore contributes zero scientific units rather than a null result. Appendix E reports execution accounting and separates scientific non-passes from support/runtime failures.

5 Write-Time Intervention: Does the Reward Label Change Persistent Memory?

We begin with a paired intervention over released WebArena Shopping trajectories from the Salesforce fragility study. For each source, the task text and released action history are copied byte-for-byte into two ReasoningBank writer calls. One call uses the released success-reflection instruction and the paired call uses the released failure-reflection instruction. DeepSeek-V4-Flash writes memory at temperature zero; task, trajectory, model, sampling, and output format stay fixed.

Original paired intervention. Four of six requested trajectory pairs return complete memory outputs in both conditions. The two incomplete requests terminate in the failure-label writer before assistant text is produced and are excluded from the paired calculation. Among the four complete pairs, every pair changes normalized memory content and extracted memory-item titles. Token-set Jaccard distance is 0.691, 0.708, 0.770, and 0.770 for tasks 21, 22, 23, and 25, respectively (mean 0.735). The 4/4 result is therefore an existence result conditional on paired completion. The content changes are procedural as well as lexical: a fixed operation-slot taxonomy changes in all four pairs (mean slot-set distance 0.493). For example, task 21's success-conditioned memory emphasizes entering the review section and preserving direct quotations, whereas its failure-conditioned memory emphasizes exhaustive pagination and verification. This structural diagnostic is descriptive; Section 6 provides the stronger reward-preserving wording reduction.

Breadth replication over 16 fresh sources. The small original source count is a natural external-validity concern, so we preregister a disjoint breadth sample before new writer outcomes. The deterministic selection balances eight originally successful and eight originally failed trajectories and first maximizes distinct intent-template families before filling by task ID; it covers 11 template families. Tasks used by the original intervention, prompt control, forced terminal test, and the first retrieval audit's held-out hits are excluded. The parent 2,200-token execution completes 29/32 writer calls but length-censors three failure-branch outputs, so it has no breadth-gate authority. A single execution-only repair raises the output cap to 4,096 uniformly and regenerates all 32 units fresh. All calls then complete with zero provider failures.

All 16 fresh pairs change exact normalized content and title sets. Their mean token-set Jaccard distance is 0.658 (range 0.545–0.850). Combined with the original four complete pairs, the write-time evidence now contains 20/20 divergent paired sources, with pooled mean token-set distance 0.673.

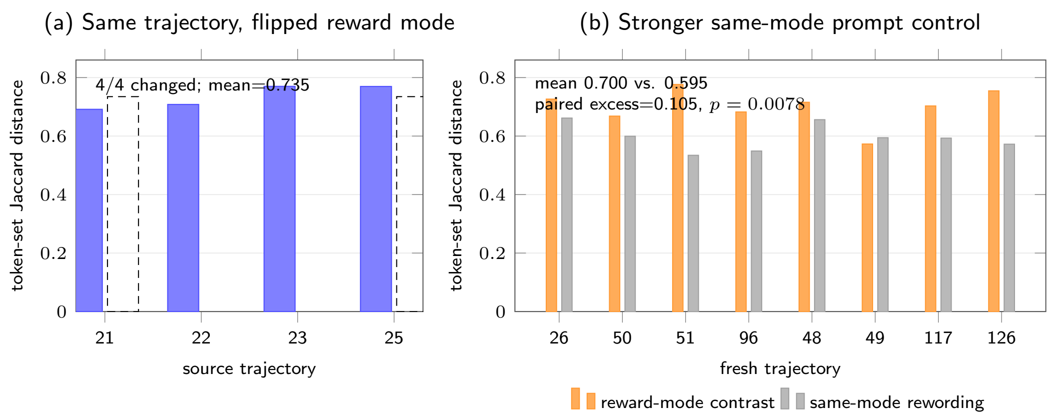

Figure 2: Write-channel evidence and its strongest simple control. (a) Among the four paired write interventions complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

Because the breadth sample is outcome-balanced and fixed before these writer outputs, this result strengthens the upstream conclusion without selecting sources for large downstream effects. Figure 2 retains the original pair/control visualization; Appendix D reports the expanded source set and execution receipt.

The write-time conclusion is therefore the strongest part of the evidence chain: under a fixed construction policy, changing which released reward-conditioned branch processes the same experience reliably changes the durable textual state. The next section asks whether this exceeds generic prompt sensitivity; Section 7 then separates forced downstream sensitivity from realized transport through the released retriever.

6 Ruling Out Generic Prompt Sensitivity

The reward intervention changes the reflection instruction together with the reward semantics. A generic prompt-sensitivity account therefore predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence. We test that reduction on eight released trajectories disjoint from the paired write sample and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\text{prompt}} = \frac{1}{8}\sum_{i=1}^{8}(B_i - W_i). \tag{7}$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

The same conclusion is visible at the procedural-structure level. Without changing the operation-slot taxonomy used in the paired write intervention, we apply it to the frozen title sets from all eight control trajectories. Mean slot-set distance between the released success and failure modes is 0.556, compared with 0.247 for semantic-preserving rewordings within the same mode, an average structural excess of 0.309. Five tasks have positive excess, two tie, and one is negative. We treat this as a descriptive controlled diagnostic, not a second significance test or an embedding-semantic claim. Its role is narrower: the structural shift associated with reward mode is larger on average than the structural shift produced by reward-preserving wording changes under the same fixed taxonomy.

7 From Durable State Change to Policy Uptake

A different memory string is not yet a different policy. We therefore trace the same reward-label intervention through four distinct gates: durable-state construction, native retrieval exposure, branch-specific policy uptake, and terminal outcome. This ordering is central to the claim: a positive forced intervention can coexist with weak native transport without contradiction.

7.1 Forced injection establishes latent leverage

The original action-reach experiment holds a future browser observation fixed while swapping only the paired persistent memory. Seven of 12 aligned rollouts choose different structured next actions, which is an existence witness rather than a population effect. A stronger first-action distribution stress test has mean empirical TV distance 0.344 but permutation $p = 0.311$, so systematic action-distribution separation is not established.

The terminal forced memory-swap test fully crosses the four original complete source-memory pairs with four future evidence packets frozen before terminal-policy calls. The initial three-rollout-per-condition run is a committed non-pass (mean $|\Delta| = 0.145833$, $p = 0.160128$). Before any confirmatory call, the follow-up freezes the identical four sources, four futures, 0.15 practical-effect floor, and $p < 0.05$ criterion; it changes only fresh replication depth from three to eight and forbids outcome-driven source or future selection. All 256 confirmatory calls complete, yielding mean $|\Delta| = 0.15625$ with permutation $p = 0.00074$. Eight of 16 cells are zero, and the two largest cells contain 78.2% of squared-effect mass. We therefore call this *forced-intervention sensitivity*: the paired states can have conditional leverage when directly injected, but this experiment bypasses native retrieval.

A fresh source-independent no-memory control on these four futures also rules out a simple "success memory helps / failure memory hurts" ordering. Omission matches both branches in eight source–future cells, lies closer to success memory in six, and closer to failure memory in two.

7.2 Native exposure is a separate gate

ReasoningBank retrieves over task descriptions with all-MiniLM-L6-v2, the current intent as query, top-1 selection, and threshold 0.3; reward-conditioned memory content is not part of the retrieval embedding. Exact replay of this released mechanism uses zero provider calls. With only the original four source descriptions, 8 of 188 held-out Shopping tasks clear the threshold (4.26%), and none of the four forced-terminal futures does. The forced 4×4 result is therefore not source-faithful retrieval transport and should not be reported as such.

After adding the 16 preregistered breadth sources, the same top-1/0.3 mechanism hits 125 of 172 held-out Shopping tasks (72.7%), spanning 39 intent-template families. Before new terminal outcomes, we freeze every hit that also has reconstructable released evidence and a deterministic evaluator. This produces 36 native tasks across 13 intent-template families and 11 retriever-selected source memories. Exposure can therefore be broad even when the branch-specific effect that follows exposure is not.

This distinction is consistent with recent trajectory-memory work that treats post-retrieval reuse as a separate problem from selecting a relevant past trajectory (Li et al., 2026b). Our setting differs in treatment—we intervene on reward-conditioned writing—but the evaluation principle is aligned: retrieval success does not determine how strongly the retrieved support changes a policy.

7.3 SHOPPING ATTENUATION APPEARS BEFORE BRANCH-SPECIFIC POLICY UPTAKE

For each of the 36 frozen native tasks, the retriever itself chooses the source. We recover exact success- and failure-conditioned memory bytes and serialize both through the released ReasoningBank human-memory wrapper. Future evidence, Doubao Seed 2.0 Mini, temperature, evaluator, and four-rollout depth are matched. All 288 calls complete without provider failure. Mean terminal S/F separation is only 0.02083 with permutation $p = 0.4289$, far below the unchanged 0.15 practical floor. Thirty-four of 36 tasks have zero branch contrast: 18 joint floors and 16 joint ceilings.

We next test three simple explanations for this attenuation. First, literal omission produces mean memory-presence effect 0.04514 ($p = 0.00147$), statistically detectable but below the 0.15 floor. Second, replacing rewritten memory with the exact compact writer-input trajectory yields mean rewrite-versus-raw effect 0.04514 ($p = 0.00775$), again below the floor; 31/36 tasks have identical S/F/raw/no-memory point estimates. Third, we hold structured rewriting itself fixed. An outcome-blind procedural writer receives the same task and action-summary bytes, the same DeepSeek-V4-Flash writer family, and the same at-most-three Title/Description/Content schema, but no success/failure, reward, score, or outcome semantics. All 20 neutral memories complete and are textually distinct from both reward branches. On the same 36 native tasks, 144/144 neutral-memory terminal calls complete with zero provider failure. The preregistered reward-conditioned-versus-neutral effect is 0.04514 with $p = 0.0048$: detectable, but again below the 0.15 floor. Thirty-two tasks have zero effect and 30/36 have identical success, failure, neutral, raw-trajectory, and no-memory point estimates.

The nonzero neutral-control effect is also highly localized. A zero-call descriptive concentration audit shows that future task 229 alone contributes 61.5% of absolute effect mass and 87.7% of squared-effect mass; removing that task lowers the mean effect from 0.04514 to 0.01786. Only two of the 11 retriever-selected source memories have any nonzero future task. This audit does not alter the preregistered permutation result or gate; it clarifies why a small p-value should not be read as broad practical transport.

Endpoint resolution is not the main explanation either. A post-hoc, zero-call fractional-reference re-score of 432 archived S/F/no-memory outcomes reduces joint S/F floors from 18 to 10, but mean S/F separation remains 0.01951 overall and 0.02827 on the 16 multi-reference tasks. Because this metric was introduced after observing saturation, it remains diagnostic-only and cannot replace the frozen binary endpoint.

Most importantly, attenuation is already visible before terminal scoring. A preregistered first-action probe freezes the released step-1 browser state on the same 36 native tasks and samples success-memory, failure-memory, and no-memory conditions with four rollouts each. All 432 calls complete with zero provider or unrecoverable parse failures. Mean S/F first-action TV is 0.06944 with permutation $p = 0.5801$, below the inherited 0.20 process floor. Only 9/36 states have any S/F TV, and 0/36 change the modal branch action. Removing click indices lowers mean S/F action-family TV to 0.02778. The larger memory-versus-no-memory TV of 0.17014 with six modal shifts is descriptive only because no confirmatory presence gate was preregistered. Zero-call post-hoc diagnostics give only 0.01659 S/F excess in next-goal text and 0.00335 pair-relative branch shift in model-emitted working memory ($p = 0.205$, $d_z = 0.148$); the latter correlates descriptively with first-action TV ($r = 0.464$) but not terminal effect ($r = -0.023$). Together, these diagnostics place the Shopping attenuation before or at *branch-specific policy uptake*; none is a confirmatory mediation test.

7.4 GENERALIZATION AND UNRESOLVED BOUNDARIES

A second-domain Reddit replication is qualified before any new writer output: eight deterministic native futures across two templates retrieve four source identities. After one uniform output-cap repair that regenerates all eight writer units fresh, all four S/F source pairs diverge (Jaccard 0.652). All 64 terminal calls complete; mean $|\Delta| = 0.125$ with $p = 0.2253$, below the unchanged 0.15 floor, with six of eight tasks zero and opposite signs in the two nonzero cells. Every leave-one-task-out mean remains below 0.15. Full qualification and repair provenance appears in Appendix D.

Cross-setting evidence therefore favors a bounded conclusion: write divergence transfers more readily than behavioral magnitude. GLM-5.3 also replicates write divergence but misses its terminal floor (0.140625, $p = 0.00012$) with two sign reversals, while the DeepSeek downstream-policy

Table 1: Stage-resolved evidence. Practical-effect floors remain binding when preregistered; a small p-value alone is insufficient.

Scientific gate	Support / calls	Key result	Verdict
Write identification	20 complete pairs	20/20 content/title divergence; pooled Jaccard .673	support
Forced leverage	16 cells / 256	mean $\|\Delta\| = .15625$, $p = .00074$	pass
Native exposure	172 held-out / 0	125 hits (72.7%); 36 terminal-eligible	support
First-action uptake	36 states / 432	S/F TV .06944, $p = .5801$; 0/36 modal diff.	non-pass
Native terminal	36 tasks / 288	S/F $\|\Delta\| = .02083$, $p = .4289$	non-pass
Outcome-blind structured control	20 writer + 144 terminal	reward-vs-neutral .04514, $p = .0048$; 32/36 zero	floor fail
Reddit cross-domain replication	4 pairs + 64 terminal	write 4/4; native $\|\Delta\| = .125$, $p = .2253$	floor fail

transfer remains a zero-scientific-unit provider-support stop rather than a scientific null. Shopping localizes attenuation before or at branch-specific uptake; Reddit shows that native magnitude itself is domain/task dependent.

8 LIMITATIONS

The original six-source intervention has two failure-branch provider incompletions, so its 4/4 estimate is conditional on paired completion. The 16/16 Shopping breadth replication and four-pair Reddit replication broaden write evidence, but Reddit terminal support remains only eight deterministic tasks across two templates. GLM-5.3 reproduces write divergence yet misses its terminal floor, so writer-invariant downstream effects remain unestablished.

Native experiments use exact released retrieval identity and the ReasoningBank wrapper but fixed browser-state packets rather than live endpoints. Shopping terminal S/F is 0.02083; no-memory, raw, and outcome-blind controls remain small, with the latter concentrated in few cells. Shopping first-action TV is 0.06944 ($p = 0.5801$), and a post-hoc working-memory audit finds only 0.00335 pair-relative branch shift ($p = 0.205$); neither is a formal mediation test. Reddit terminal S/F is descriptively larger at 0.125 ($p = 0.2253$) but still misses the common 0.15 floor. We therefore do not treat the Shopping magnitude as a cross-domain law or the Reddit point estimate as positive practical transport. DeepSeek policy transfer remains a provider-support stop. Claims remain tied to the released top-1/0.3 retrieval substrate; learned retrieval, multi-memory composition, live interaction, and population risk are outside scope.

9 CONCLUSION

A terminal evaluator can become a state updater when its label selects how an experience is committed to persistent memory. Under byte-identical experiences, reward-conditioned writing changes durable state on 20/20 complete Shopping pairs, survives a stronger wording control, and replicates on all four retriever-required Reddit source pairs. Forced injection shows latent leverage ($|\Delta| = 0.15625$, $p = 0.00074$).

Native behavior is more conditional. Shopping yields first-action TV 0.06944 and terminal $|\Delta| = 0.02083$; Reddit terminal $|\Delta| = 0.125$ is larger but also misses the same 0.15 floor. The central lesson is therefore stage-resolved and domain-bounded: *reward errors change memory before they change policy*. Memory divergence establishes persistent-state corruption, not a domain-invariant behavioral effect. Downstream risk should be audited through write, exposure, policy uptake, and outcome rather than inferred from stored text alone.

References

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A F0 Reproduction Details

A.1 Released trajectory sample

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 Memory construction intervention

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

Table 2: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

A.3 PAIR-LEVEL F0 RESULTS

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B PROMPT-PERTURBATION CONTROL DETAILS

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 3 reports the paired distances used in the frozen sign-flip test.

Table 3: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C DOWNSTREAM CONFIRMATORY DETAILS

C.1 FIXED-STATE ACTION DISTRIBUTIONS

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

C.2 FROZEN TERMINAL MATRIX

The initial terminal experiment and F2R1 use the same complete F0 source set $\{21, 22, 23, 25\}$, the same future-task set $\{164, 385, 387, 388\}$, and the same aggregate absolute-effect statistic with the same dual gate: mean absolute difference at least 0.15 and within-cell permutation $p < 0.05$.

The initial stage uses three fresh rollouts per condition per cell and is already frozen in Git commit `730c2bc` at 2026-08-22 00:01:42 +08:00; it yields 0.145833 with $p = 0.160128$ and does not pass. The separately authorized F2R1 contract is content-addressed as `ea9a2260...01f5`; its first provider receipt is generated at 2026-08-22 01:50:32 +08:00. It explicitly forbids source/future-task selection after outcomes and increases only uniform replication depth to eight fresh rollouts per condition in every cell. All 256 F2R1 calls complete, yielding 0.15625 with $p = 0.00074$. Thus F2R1 is a targeted same-support replication after an initial non-pass, not a claim that the confirmatory design was chosen before any terminal outcomes were observed.

Table 4: Initial F2 and F2R1 cell-level terminal rates on identical 4-by-4 support. S/F denote success-/failure-label memory.

Source	Future	F2 S	F2 F	F2 $\|\Delta\|$	F2R1 S	F2R1 F	F2R1 $\|\Delta\|$
21	164	1.000	1.000	0.000	1.000	1.000	0.000
21	385	0.000	0.333	0.333	0.000	0.250	0.250
21	387	0.000	0.333	0.333	0.125	0.250	0.125
21	388	1.000	0.333	0.667	1.000	0.625	0.375
22	164	1.000	1.000	0.000	1.000	1.000	0.000
22	385	0.000	0.667	0.667	0.000	0.125	0.125
22	387	0.000	0.000	0.000	0.125	0.000	0.125
22	388	1.000	1.000	0.000	0.250	1.000	0.750
23	164	1.000	1.000	0.000	1.000	1.000	0.000
23	385	0.000	0.000	0.000	0.000	0.000	0.000
23	387	0.000	0.000	0.000	0.125	0.250	0.125
23	388	0.667	1.000	0.333	1.000	1.000	0.000
25	164	1.000	1.000	0.000	1.000	1.000	0.000
25	385	0.000	0.000	0.000	0.000	0.000	0.000
25	387	0.000	0.000	0.000	0.125	0.750	0.625
25	388	1.000	1.000	0.000	1.000	1.000	0.000
Mean $\|\Delta\|$		0.145833			0.15625		

C.3 Fresh no-memory branch-location control

The no-memory control is frozen after F2R1 as a secondary branch-location diagnostic. It uses the identical four future tasks, released evidence packets, Doubao Seed 2.0 Mini settings, evaluator, and eight-rollout depth, but no source-memory dimension. The earlier 12 exploratory no-memory calls are excluded. An initial 32-call execution is also excluded from science because a new local validator incorrectly required the provider's resolved model name to equal the requested alias; all 32 responses resolved to `doubao-seed-2-0-mini-260215`, which is exactly the versioned model name recorded in all 256 historical F2R1 receipts. After freezing that execution-layer diagnosis and archiving provider responses before local validation, the one-for-one recovery completes 32/32 calls. Both attempts are counted in execution cost; only the recovery outcomes enter the analysis.

Table 5: Fresh no-memory terminal baseline. Wilson intervals are per future task and are not replicated across source memories.

Future task	Successes	Rate	Wilson 95% interval
164	8/8	1.000	[0.676, 1.000]
385	0/8	0.000	[0.000, 0.324]
387	0/8	0.000	[0.000, 0.324]
388	8/8	1.000	[0.676, 1.000]

Combining these four source-independent baselines descriptively with the 16 F2R1 cells yields eight cells where omission matches both memory branches, six where omission is closer to the success-memory branch, and two where it is closer to the failure-memory branch. The largest illustrative contrasts are source/future 22/388, where $(p_S, p_F, p_0) = (0.25, 1.0, 1.0)$, and 25/387, where $(p_S, p_F, p_0) = (0.125, 0.75, 0.0)$. The first isolates the success-conditioned memory as the deviation from omission; the second places both memories above omission. Because each p_0 is shared by four source rows, we report cellwise score intervals and no new global significance test.

C.4 Cross-writer replication and frozen negative boundary

The writer-family test is sequential. Stage 1 first rewrites only source tasks 21, 22, 23, and 25 under the same released success/failure ReasoningBank prompts using GLM-5.3 at temperature zero. The parent 2,200-token attempt is scientifically invalid after two failure-mode responses terminate at the exact output cap with no assistant text; an overlapping-runner race also makes its provider POST count reconstructible only as a lower bound of nine. The single preregistered repair changes only the output cap to 4,096 and adds a single-writer transaction lock. Its eight calls all complete; all four pairs change content and title sets, with token-set distances 0.783, 0.728, 0.715, and 0.723 (mean 0.737482).

Stage 2 keeps the original Doubao Seed 2.0 Mini downstream policy, four future tasks, evidence packets, evaluator, eight-rollout cell depth, and dual terminal gate unchanged. All 256 cells-by-condition rollouts are scorable. The observed mean absolute effect is 0.140625 and the 100,000-draw permutation test gives $p = 0.00012$. The statistical gate passes but the frozen 0.15 minimum-effect gate does not, so the registered cross-writer terminal generalization claim is not established. Table 6 shows that the difference is not a uniform attenuation.

Table 6: Signed failure-minus-success terminal effects under the original DeepSeek-V4-Flash writer (D) and the GLM-5.3 writer (G).

Source	Future	D effect	G effect
21	164	0.000	0.000
21	385	0.250	0.250
21	387	0.125	0.375
21	388	-0.375	0.000
22	164	0.000	0.000
22	385	0.125	0.000
22	387	-0.125	-0.125
22	388	0.750	-0.250
23	164	0.000	0.000
23	385	0.000	0.000
23	387	0.125	-0.750
23	388	0.000	0.000
25	164	0.000	0.000
25	385	0.000	0.000
25	387	0.625	0.500
25	388	0.000	0.000

Among the six cells that are nonzero under both writers, four retain direction and two reverse. Reusing the already frozen writer-independent no-memory rates only as descriptive baselines yields 10 GLM cells equidistant from omission, four closer to the success-memory branch, and two closer to the failure-memory branch; no additional provider calls or global test are introduced. This supports the upstream write-channel replication while preserving the failed terminal-invariance gate.

C.5 Heterogeneity and corruption-rate decomposition

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the paper treats the result as a variance channel instead of a uniform directional harm effect.

If reward-label corruption selects the failure-conditioned memory with probability q, the induced between-memory variance equals $q(1-q)\Delta_c^2$ in cell c. The exact mean squared conditional effect across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1-q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source–future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.

C.6 Controlled structural diagnostic

The controlled structural diagnostic reuses the exact operation-slot taxonomy already frozen for the F0 title audit; it does not add a learned semantic judge. Applied to the eight disjoint prompt-control trajectories, the mean slot-set distance between released reward modes is 0.555655, whereas the mean distance induced by semantic-preserving rewording within the same mode is 0.246578. The mean between-minus-within structural excess is 0.309077; five tasks are positive, two tie, and one is negative. This is descriptive supporting evidence rather than a second hypothesis test. On the four complete F0 pairs, success-only operation slots include at least one of evidence capture, navigation, or query expansion in all four pairs, while verification/completeness is failure-only in three pairs. Provider receipts also show longer failure-mode responses in all four complete pairs (mean output-token difference +266.5; mean reasoning-token difference +268.0), which we report only as response-structure asymmetry, not as semantic quality.

C.7 Source–future interaction diagnostic

A two-way additive decomposition of the centered signed F2R1 cell effects is shown below. It is a finite-matrix description, not inferential ANOVA.

Table 7: Descriptive decomposition of centered signed terminal effects.

Component	Sum of squares	Share
Source-memory main effect	0.101562	9.4%
Future-task main effect	0.070312	6.5%
Source $\times$ future interaction residual	0.906250	84.1%
Total	1.078125	100%

The source and future marginals therefore summarize little of the signed effect structure. Source 21 is positive on future tasks 385 and 387 but negative on 388; source 22 is positive on 385 and 388 but negative on 387. Future tasks 387 and 388 likewise change sign across source memories. Table 8 adds benchmark-observable future-task attributes. Future task 164 is at a joint 1.0/1.0 terminal ceiling for every source pair, so it has no observable binary-outcome headroom and contributes zero effect mass. The other three tasks contain all observed squared-effect mass. With only four future tasks, start context, task identity, and outcome headroom are confounded; these rows characterize where the frozen effect occurred but do not define a general transfer predictor.

Table 8: Future-task attributes and concentration of the frozen terminal effect. "Home" means the task starts at the Shopping root rather than a product page.

Task	Start	Ref. items	Mean $\lvert\Delta\rvert$	Squared-mass share
164	Product	2	0.00000	0.0%
385	Home	8	0.09375	6.4%
387	Home	6	0.25000	35.9%
388	Home	2	0.28125	57.7%

A second descriptive check asks whether sources whose paired memories differ more at write time simply create larger downstream effects. Table 9 shows that this reduction is not supported by the four completed sources. Sources 23 and 25 are effectively matched on both reported write-divergence magnitudes but differ fivefold in source-averaged terminal sensitivity. Across the four points, the descriptive Pearson correlations are -0.730 for token distance and -0.367 for slot distance versus mean terminal $|\Delta|$. With $n = 4$ these correlations have no inferential status; they are included only to reject the simple monotonic reading that a larger textual or operation-slot perturbation necessarily produces a larger downstream effect.

Table 9: Write-stage divergence versus source-averaged terminal sensitivity. The final column averages $|\Delta|$ over the four frozen future tasks.

Source	Token distance	Slot distance	Mean terminal $|\Delta|$
21	0.691	0.571	0.188
22	0.708	0.400	0.250
23	0.770	0.500	0.031
25	0.770	0.500	0.156

D Breadth, Retrieval, and Transport Expansion

D.1 Sixteen-source write replication

The breadth sample is selected before new writer outputs. It excludes all original write/control tasks, the four forced-terminal futures, and the eight held-out tasks identified as retrieval hits by the first exact-retriever audit. Within each original-outcome stratum, selection first takes the lowest task ID from each previously unused intent-template family, then fills remaining slots by task ID. Table 10 reports all 16 fresh pairs. The 2,200-token parent run produces three failure-branch length-censored calls (tasks 118, 125, and 232); its 29 completed units are not used for the breadth gate. The sole repair changes only the output cap to 4,096, regenerates all 32 units fresh, and completes every pair.

Table 10: Fresh source-breadth replication. O is the trajectory's original benchmark outcome; F/S denote failure/success. All 16 pairs change both exact content and memory-item title sets.

Task	O	Jaccard	Task	O	Jaccard
118	F	.850	145	S	.594
125	F	.693	148	S	.594
141	F	.678	159	S	.732
146	F	.565	188	S	.682
158	F	.668	225	S	.587
163	F	.642	226	S	.732
189	F	.699	231	S	.545
232	F	.550	260	S	.712

The fresh mean token-set distance is 0.65757. Pooling these 16 complete pairs with the four original complete pairs gives 20/20 exact-content changes, 20/20 title-set changes, and pooled mean token-set distance 0.673014.

D.2 Exact released-retriever exposure

The retrieval audit directly executes the cached all-MiniLM-L6-v2 encoder with the released SentenceTransformer mean-pooling and normalization configuration, maximum sequence length 256, top-1 retrieval, and threshold 0.3. No provider call is used. Table 11 separates exposure under the original sparse bank from exposure after source expansion.

Table 11: Exact released-retriever exposure on held-out Shopping tasks. "Eligible" additionally requires a released trajectory and deterministic offline string-match evaluator.

Memory bank	Held-out	Hits	Hit rate	Eligible
4 original sources	188	8	4.26%	–
20 sources	172	125	72.67%	36

None of the original four forced-terminal futures retrieves an original-bank memory. After expansion, future task 164 retrieves new source 163 with cosine 1.0; the other three original futures remain below threshold. Across the 20-source bank, the 125 hits span 39 intent-template families; the 36 pre-outcome terminal-eligible tasks span 13 families and retrieve 11 distinct sources.

D.3 FULL 36-TASK NATIVE-WRAPPER OUTCOMES

Table 12 exposes all frozen transport cells. S/F are success- and failure-conditioned retrieved-memory rates; N is the independently generated no-memory rate. Each rate uses four fresh Doubao Seed 2.0 Mini rollouts. The reward-branch experiment has mean $|S - F| = 0.020833$ and permutation $p = 0.428866$. The three-arm memory-presence statistic has mean 0.045139 and omnibus $p = 0.00147$, but misses the frozen 0.15 practical floor. Only tasks 144, 228, and 319 have nonzero branch-to-omission magnitude; 34/36 tasks have S=F, comprising 18 joint 0/0 floors and 16 joint 1/1 ceilings.

Table 12: All 36 pre-outcome retrieval-matched fixed-evidence tasks. Src is the exact released top-1 source. P is the mean absolute memory-presence effect $\frac{1}{2}(|S - N| + |F - N|)$.

Task	Src	S	F	N	P	Task	Src	S	F	N	P
26	25	0	0	0	0	229	226	1	1	1	0
124	226	0	0	0	0	230	226	1	1	1	0
126	226	1	1	1	0	233	232	1	1	1	0
142	141	0	0	0	0	279	226	0	0	0	0
143	141	0	0	0	0	280	226	0	0	0	0
144	141	.75	1	0	.875	281	225	0	0	0	0
147	146	0	0	0	0	282	226	0	0	0	0
149	148	1	1	1	0	319	188	1	1	.5	.5
150	226	1	1	1	0	320	188	0	0	0	0
164	163	1	1	1	0	321	188	0	0	0	0
165	163	0	0	0	0	322	188	1	1	1	0
167	163	0	0	0	0	323	188	0	0	0	0
190	189	1	1	1	0	329	141	1	1	1	0
192	188	0	0	0	0	330	141	0	0	0	0
227	226	1	1	1	0	331	141	1	1	1	0
228	226	.5	0	0	.25	333	141	0	0	0	0
358	189	1	1	1	0	360	231	1	1	1	0
362	231	1	1	1	0	384	25	0	0	0	0

D.4 RAW WRITER-INPUT TRAJECTORY BASELINE

A stronger native-support comparator replaces the rewritten memory with the compact action/result trace that the C1 writer itself receives before counterfactual reward-label conditioning. Source identity is still selected by the exact released top-1/0.3 retriever. No counterfactual success/failure label and no rewritten memory item are supplied. Future evidence, Doubao Seed 2.0 Mini, temperature, evaluator, and four-rollout depth are identical to B4/B5. All 144 calls complete with zero provider failures. The preregistered statistic $\frac{1}{2}(|S - R| + |F - R|)$ averages 0.045139; a 100,000-repetition three-arm permutation test gives $p = 0.00775$, but the unchanged 0.15 practical floor is missed. This is therefore a statistical deviation without a practically large rewrite-versus-raw effect on the frozen support.

Thirty-one of 36 tasks have identical success-memory, failure-memory, raw-trajectory, and no-memory point estimates, and raw equals no-memory on 32/36. Table 13 lists the only five tasks where raw differs from at least one other arm. A derived tie-aware audit is used for geometry only: 34/36 tasks place raw equally close to success memory, failure memory, and no-memory; task 144 is uniquely closest to failure memory, and task 228 ties failure memory with no-memory. We exclude the runner's single-label "closest arm" secondary field because its deterministic lexical tie-break collapses these ties; this reporting correction does not affect the primary B8 statistic, permutation test, or verdict.

Table 13: The five native-support tasks where the raw writer-input trajectory rate R differs from at least one other point estimate. S/F are reward-conditioned memories and N is omission.

Task	Src	S	F	R	N
144	141	.75	1	1	0
228	226	.5	0	0	0
230	226	1	1	.5	1
319	188	1	1	.75	.5
331	141	1	1	.5	1

D.5 POST-HOC ENDPOINT-HEADROOM DIAGNOSTIC

Sixteen of the 36 native-support tasks require more than one reference substring. The released binary evaluator multiplies these checks, so missing any required item maps the rollout to zero. After observing B4/B5 saturation, we therefore re-score the already archived 432 success/failure/no-memory outputs with fractional reference coverage. This introduces zero provider calls and is explicitly diagnostic-only: it has no confirmatory gate, does not alter the released evaluator, and cannot retroactively rescue B4/B5.

The diagnostic confirms some endpoint compression but not a hidden large reward-branch effect. Joint success/failure floors fall from 18 under the binary score to 10 under fractional coverage. Nevertheless, mean absolute S/F separation is 0.019511 over all 36 tasks and 0.028274 over the frozen 16-task multi-reference subset. Mean absolute memory-presence separation under fractional coverage is 0.045635 over all 36 tasks. Only three tasks have identical binary S/F means but different fractional-coverage S/F means. We therefore retain endpoint headroom as one transport gate without using score resolution to explain away the native branch non-pass.

D.6 NATIVE FIRST-ACTION TRANSPORT LOCALIZATION

To test whether terminal attenuation occurs only after decisions are made, we freeze a pre-terminal process endpoint on the identical 36 native-hit tasks. For every task, the selected source remains the exact released top-1 source. The future observation is the released trajectory's step-1 browser state, with any pre-existing task-history memory stripped before the experiment; there is no task-specific step selection. Success-memory, failure-memory, and no-memory conditions each receive four fresh Doubao Seed 2.0 Mini rollouts at the same temperature and output cap as B4/B5. The action signature is inherited from the earlier F1D protocol: first structured action name, with click-element index retained for clicks. All 432 calls complete with zero provider or unrecoverable parse failures.

The preregistered statistic is mean per-state empirical TV between success- and failure-memory first-action distributions, with the inherited 0.20 process floor and 100,000-draw within-state permutation test. The observed mean TV is 0.069444 with $p = 0.580094$, so branch-specific first-action transport is not established. Nine of 36 states have any nonzero S/F TV, but none changes its modal S/F action. Coarsening clicks by removing element indices reduces mean S/F action-family TV to 0.027778 and leaves only four states with any family-level difference. The secondary no-memory geometry is larger but descriptive: mean $\frac{1}{2}(\mathrm{TV}(S,N) + \mathrm{TV}(F,N)) = 0.170139$, and six states have a modal action under at least one memory branch that differs from no-memory. Because no confirmatory presence gate was preregistered, this supports only the interpretation that generic procedural context can influence action selection while reward-branch-specific uptake remains weak.

A zero-call post-hoc next-goal diagnostic uses the archived B10 outputs without adding provider calls. Mean cross-branch token-set Jaccard distance between success and failure next-goal texts is 0.464370, while the average within-branch rollout distance is already 0.447776; the branch-specific excess is therefore only 0.016593. By contrast, mean memory-versus-no-memory next-goal distance is 0.554403. This diagnostic has no inferential authority and cannot rescue the B10 gate; it further localizes the observed attenuation before or at branch-specific operative planning rather than solely at terminal scoring.

Table 14: B10 process localization on the 36 frozen native-hit states. Presence TV is $\frac{1}{2}(\mathrm{TV}(S,N) + \mathrm{TV}(F,N))$.

Metric	Value	Registered status
Mean S/F first-action TV	0.069444	floor fail (0.20)
Permutation p	0.580094	fail
States with nonzero S/F TV	9/36	descriptive
States with modal S/F action difference	0/36	descriptive
Mean coarse action-family S/F TV	0.027778	post-hoc
Mean memory-presence first-action TV	0.170139	secondary
States with memory/no-memory modal shift	6/36	secondary

D.7 POST-HOC WORKING-MEMORY BRANCH ATTRIBUTION

A zero-call diagnostic reuses the 432 archived first-action outputs. The response schema emits a `current_state.memory` field before the first structured action; it is recoverable for all 432 outputs (405 strict JSON, 27 narrow string recoveries). We embed this text and the exact S/F input memories with the same cached all-MiniLM-L6-v2 encoder. For output y, the pair-relative margin is $m(y) = \cos(E(y), E(M_S)) - \cos(E(y), E(M_F))$. Because this observable was introduced after the first-action experiment, every quantity here is post-hoc and has no confirmatory gate.

Mean task-level branch shift is 0.003347 (median 0.003835), with 21 positive and 15 negative tasks, paired $d_z = 0.148$, and within-state label-permutation $p = 0.2052$. A common-centroid diagnostic is larger at 0.02233 but remains suggestive only ($d_z = 0.258$, sign-flip $p = 0.0664$). Branch-relative working-memory shift correlates descriptively with first-action TV (Pearson $r = 0.464$, Spearman $\rho = 0.419$; leave-one-out Pearson range 0.358–0.515) but not terminal absolute effect ($r = -0.023$). Input S/F memory cosine distance averages 0.131 and correlates only 0.095, 0.115, and 0.137 with working-memory shift, first-action TV, and terminal effect, respectively. We use this layer only to localize the Shopping boundary and explicitly forbid selecting positive cells for another provider experiment.

Table 15: Zero-call working-memory localization. All quantities are post-hoc diagnostics.

Metric	Value
Pair-relative branch shift	0.003347
Within-state permutation p	0.2052
Paired effect size d_z	0.148
Positive / negative task shifts	21 / 15
Common-centroid uptake	0.02233
Common-centroid sign-flip p	0.0664
Working-memory shift vs. first-action TV	$r = 0.464$
Working-memory shift vs. terminal effect	$r = -0.023$

D.8 OUTCOME-BLIND STRUCTURED-MEMORY CONTROL

The strongest same-information representation control removes outcome semantics while retaining procedural rewriting. Before any B11 provider call, we freeze all 20 source trajectories, a neutral prompt, and the existing 36-task native terminal support. The neutral writer receives the same task prompt and exact deterministic action-summary bytes as the reward-conditioned writer, uses DeepSeek-V4-Flash at temperature zero, and returns at most three memory items in the same Title/Description/Content schema. A literal prompt audit confirms that the neutral instruction contains no success/failure, reward, score, or outcome semantics. All 20 writer calls complete with zero provider failures; all 11 sources required by native retrieval are present before the terminal stage activates.

The neutral outputs are not copies of either reward branch. Across 20 sources, neutral title sets equal neither success nor failure title sets. Mean token-set Jaccard distance is 0.611790 from success memory and 0.690062 from failure memory; neutral is textually closer to success on 16 sources and to failure on four. These distances are descriptive only and do not define the terminal gate.

Each native task then receives the neutral memory from the same retriever-selected source, serialized through the same released ReasoningBank wrapper used by the success/failure arms. Future evidence, Doubao Seed 2.0 Mini, temperature, evaluator, and four-rollout depth remain fixed. All 144 terminal calls complete. The preregistered statistic $\frac{1}{2}(|S-U|+|F-U|)$, where U denotes the outcome-blind structured memory rate, averages 0.045139 with permutation $p = 0.0048$. Because the unchanged practical floor is 0.15, the joint gate fails. Thirty-two of 36 tasks have zero reward-conditioned-versus-neutral effect; neutral equals success on 33 tasks, failure on 33, no-memory on 32, and raw trajectory on 31. Thirty of 36 tasks have identical point estimates across all five arms.

A zero-call concentration audit clarifies the small-p, small-effect geometry. Future task 229 contributes 61.5% of absolute effect mass and 87.7% of squared-effect mass; the top two cells contribute 93.2% of squared-effect mass. Removing task 229 reduces mean effect to 0.017857. Only sources 141 and 226—2 of the 11 retriever-selected sources—contain any nonzero future task. This is descriptive robustness evidence and does not change the preregistered permutation result or floor-fail verdict.

For restartability and audit, the run retains response-first provider archives and per-call stage JSONs. We additionally materialize deterministic zero-call CSV projections from the frozen results: 20 writer rows, 144 terminal-rollout rows, and 36 cell rows. The CSV files are projections rather than replacement scientific units; their SHAs are bound in the concentration artifact.

Table 16: Outcome-blind structured-memory control on the frozen native support. The detected statistical difference remains below the preregistered practical floor and is concentrated in a few cells.

Metric	Value
Neutral writer calls	20/20 complete
Neutral terminal calls	144/144 complete
Mean reward-conditioned vs. neutral effect	0.045139
Permutation p	0.0048
Practical floor	0.15 (fail)
Zero-effect tasks	32/36
All five arms equal	30/36
Task 229 absolute-effect mass share	61.5%
Leave-task-229-out mean effect	0.017857
Sources with any nonzero task	2/11

D.9 Cross-domain Reddit replication

A zero-call qualification freezes the second-domain support before any new writer output. We start from the released AWM shuffle-1 Reddit trajectory file (106 trajectories) and all 129 Reddit task configs. All released string-match tasks are reserved from source selection. Within each original-outcome stratum, a deterministic rule first maximizes intent-template diversity and then fills by task ID, producing 20 source descriptions balanced 10 successful / 10 failed across 15 templates. Exact all-MiniLM-L6-v2 top-1/0.3 replay yields 102 held-out retrieval hits. Eight tasks also have released trajectories and deterministic offline string-match evaluators, spanning two intent templates and retrieving four distinct source identities (404, 405, 595, 610). This passes the preregistered support gate of at least six eligible hits, two templates, and two selected sources with no provider call.

Only those four retriever-required sources enter the writer replication. The parent freezes eight success/failure writer calls at a 4,096-token cap; seven complete, while source 610's failure branch length-censors without assistant text. No parent success is reused scientifically. A single execution-only repair raises only the writer output cap to 8,192 and regenerates all eight units fresh; a second repair is forbidden. All repaired calls complete. All four pairs change both content and title sets, with mean token-set Jaccard distance 0.652342.

The repaired memories are serialized through the same released ReasoningBank wrapper. On all eight qualified futures, future evidence, Doubao Seed 2.0 Mini, evaluator, temperature, and four-rollout depth are matched across S/F branches. All 64 calls complete with zero provider failures. Mean $|S - F| = 0.125$ and the 100,000-draw within-task permutation test gives $p = 0.225268$, so the unchanged 0.15/$p < 0.05$ gate fails. Six tasks are zero. Task 29 has signed $F - S = +0.5$ and task 67 has $F - S = -0.5$; they lie in different intent templates, so the aggregate signed mean is exactly zero. Every leave-one-task-out mean remains below 0.15 (range 0.071429–0.142857). This supports cross-domain write divergence while retaining a bounded, heterogeneous native-transport non-pass.

D.10 Cross-policy support stop

The downstream-policy transfer is frozen on the identical 36-task support before any DeepSeek-V4-Flash outcome. At the Doubao-matched 900-token cap, the first frozen unit ends with provider status incomplete, reason `length`, and no assistant text. One execution-only repair raises the cap uniformly to 2,200, requires all 288 units to be fresh, and forbids a second repair; the same first unit again length-censors without assistant text. The cross-policy line therefore stops after two provider POSTs with zero scientific units. The preregistered no-memory continuation is not triggered because its condition is complete paired execution, regardless of paired scientific outcome.

Table 17: Second-domain Reddit replication. Writer divergence transfers; native terminal magnitude is larger than Shopping but still misses the unchanged practical/statistical gate.

Metric	Value
Zero-call qualified future tasks	8
Qualified intent templates / selected sources	2 / 4
Fresh repaired writer pairs	4/4 divergent
Writer mean token Jaccard distance	0.652342
Scientific terminal calls	64/64 complete
Mean native $\|S - F\|$	0.125
Permutation p	0.225268
Practical floor	0.15 (fail)
Zero-effect tasks	6/8
Leave-one-task-out mean range	0.071429–0.142857

D.11 Bounded corruption-rate consequence

For completeness, the earlier forced-swap matrix implies a descriptive between-memory variance under a hypothetical reward-label corruption probability q. If $p_S(c)$ and $p_F(c)$ are the forced-swap conditional success probabilities in cell c, then $V_c(q) = q(1-q)[p_F(c) - p_S(c)]^2$. The mean squared conditional effect across the 16 cells is 0.076172, giving $\bar{V}(q) = 0.076172q(1-q)$ and 0.019043 at $q = 0.5$. A 100,000-repetition cell bootstrap gives a descriptive $V(0.5)$ interval $[0.002686, 0.041016]$. This is a deterministic consequence analysis of the forced intervention, not an empirical corruption sweep or a claim about native transport.

E Execution Accounting and Reliability

All scientific evidence is inference-only; there are no training runs or local GPU fine-tuning runs. Before the baseline-aligned expansion, the observable provider-POST lower bound was at least 841, excluding the unresolved low-level request count of the early 12-unit aligned action witness. The original expansion adds 498 observable POSTs. The raw writer-input follow-up adds 144 terminal POSTs, native first-action localization adds 432 process POSTs, the outcome-blind structured-memory control adds 20 writer plus 144 terminal POSTs, and the Reddit cross-domain program adds 80 observable POSTs (8 execution-parent writer POSTs plus 8 fresh repaired writer and 64 terminal POSTs). Across these expansion/follow-up stages this is 1,318 observable POSTs, of which 1,276 are scientifically usable completions: 60 writer calls, 784 terminal rollouts, and 432 first-action rollouts. The updated full-paper observable POST lower bound is therefore at least 2,159. Retrieval audits, endpoint-headroom and tie-aware analyses, the first-action next-goal diagnostic, working-memory attribution, B11 concentration/CSV projections, and Reddit support qualification use zero provider calls.

The 32-call Shopping breadth parent completes 29 units but length-censors three failure-branch writes; only its fresh uniform 4,096-token repair has breadth authority. The Reddit writer parent similarly completes 7/8 but length-censors source 610's failure branch at 4,096; the only authorized repair regenerates all eight writer units fresh at 8,192 and then completes 64/64 terminal calls. Parent writer successes are excluded from the Reddit scientific result. Native branch, no-memory, raw-trajectory, first-action, outcome-blind structured-memory, and Reddit R1 scientific stages otherwise complete without provider failure; the first-action stage also has zero unrecoverable parse failures. The DeepSeek policy-transfer parent and sole output-cap repair each fail on their first unit with no assistant text and contribute zero scientific units. Earlier execution failures—the two original F0 failure-arm incompletions, the execution-invalid first no-memory attempt, and the failed parent GLM writer attempt—remain recorded separately and are never imputed as scientific outcomes. Provider outputs are content-addressed or response-first archived where the frozen runner supports it.